Vinnakota Hema Kasi Sainath

AI Engineer • Python • Generative AI
+91 9133839569 | kasisainath14@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Results-driven AI Engineer with 3.5 years of experience specializing in Generative AI and Python backend development. Deep expertise in RAG pipelines, Agentic AI frameworks, multi-agent orchestration, LLMs, and MCP integrations. Proven ability to rapidly turn PoCs into production-grade solutions. Experienced building scalable RESTful APIs and microservices with FastAPI, Docker, Celery, and deploying AI systems on AWS.

TECHNICAL SKILLS

AI / ML: RAG Pipelines, Agentic AI, Multi-Agent Orchestration, LangChain, Agno, RAGAS, MCP (Model Context Protocol), A2A, LLMs (Claude, Gemini, OpenAI, GPT-4), Voice Agents (Pipecat, Deepgram, Cartesia, Sarvam), AWS Bedrock, AgentCore, Strands

Backend: Python (Async/Sync), FastAPI, Celery, Docker, Kafka, SQLAlchemy, SQLModel, Pydantic, asyncio, Pytest

Databases: PostgreSQL, MySQL, Neo4j, Vector DB, Supabase, OpenSearch Serverless

AI Dev Tools: Streamlit, Gradio, Cursor, Windsurf, Kiro, Postman, PgAdmin, DBeaver, Git, JIRA

OCR / Document: Whisper (LLM-based Markdown OCR), EasyOCR, Tesseract, SharePoint document integration

PROFESSIONAL EXPERIENCE

Software Engineer I | **Smart IMS (Hyderabad)** *Dec 2022 – Present*

- Specialised as an AI Engineer building production RAG pipelines, agentic workflows, and LLM-powered backends using FastAPI and Python.
- Built end-to-end Chat Agent platform using the Agno agentic framework, enabling users to configure LLMs, tools, guardrails, MCPs, and RAG-based knowledge integration — converting quick PoCs into working solutions.
- Designed and deployed RAG-based document processing APIs: extraction, chunking, and summarisation to power 'Chat with Doc' and notebook-style LLM features.
- Implemented real-time conversational voice agents using Pipecat with Deepgram & Cartesia (English) and Sarvam (Indian regional languages) for STT/TTS — not TTS-only, but full conversational audio agents over WebSockets and WebRTC.
- Integrated AWS Bedrock, AgentCore, and Strands to build cloud-native AI agent infrastructure for enterprise clients.
- Integrated LLMWhispher OCR for converting scanned insurance documents into clean Markdown, combined with EasyOCR and Tesseract for hybrid extraction across document types.
- Implemented JWT authentication and Role-Based Access Control (RBAC) for secure multi-user AI applications.
- Leveraged Celery and FastAPI Background Tasks for heavy, long-running AI/ML workflows and batch processing pipelines.
- Collaborated in Agile sprints with cross-functional teams to deliver AI-driven enterprise solutions.

PROJECTS

AI Agent for Insurance Carrier – Coaction (AWS-Native)

- Built AI agent on AWS for Coaction, an insurance carrier, leveraging AWS Bedrock, AgentCore, and the Strands agent framework for orchestration and tool use.
- Built a multi-source RAG pipeline ingesting SharePoint documents and web-scraped data, enabling agents to answer complex insurance queries with accurate, grounded responses.
- Implemented intelligent web scraping modules (Crawl4AI) to continuously update the knowledge base with carrier-specific and regulatory information.
- Designed the system for scalability on AWS (S3, Lambda, Bedrock) with secure access controls, enabling the agent to serve underwriters and ops teams through a Streamlit/Gradio interface.

Tech Stack: *AWS Bedrock, AgentCore, Strands, RAG, SharePoint , Web Scraping, S3, Lambda, Python, FastAPI, Vector DB, OpenSearch serverless*

Xymphony AI – Agent Development Platform

- Developed a scalable backend using FastAPI and Python integrating multiple LLM providers (OpenAI, Google Gemini, Anthropic Claude) via a unified API interface.
- Built modular agent architecture with the Agno framework, supporting custom tools and dynamic RAG knowledge retrieval — enabling rapid PoC-to-production cycles.
- Implemented a RAG evaluation system using RAGAS and custom metrics to continuously improve agent accuracy and retrieval quality across diverse use cases.
- Containerised with Docker and set up CI/CD pipelines; delivered Streamlit and Gradio UIs for agent testing and demos.

Tech Stack: *FastAPI, Python, Agno, RAG, RAGAS, LangChain, OpenAI, Gemini, Anthropic, Vector DB, Docker, Streamlit, Gradio*

Clinical Trial Management System (CTMS)

- Designed and developed a system to streamline planning, tracking, and management of clinical trials using microservices architecture.
- Integrated Keycloak for user authentication, token exchange, and permissions management.
- Integrated with EDC systems (ETMF and ISF) for secure document storage and trial data collection; implemented version control for trial protocols ensuring compliance.

Tech Stack: *FastAPI, Python, Keycloak, Microservices, PostgreSQL, Docker*

EDUCATION

Bachelor of Technology *Jun 2019 – Apr 2022*

DMS SVH College of Engineering, Machilipatnam | CGPA: 7.58/10

CERTIFICATIONS & ACHIEVEMENTS

- Extra Miler Award – Recognised for exceptional dedication at Smart IMS (2024).
- Difference Maker Award – Recognised for outstanding impact and contributions at Smart IMS, December 2025.